

Drosophila Brain Cocaine Response

Single-Cell RNA-Seq Analysis • Student Project Guide (Project 2)

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1. Project Timeline & 15-Day Milestone Deadlines

This project has a **15-day execution timeline**, starting **Today, August 25, 2026** and concluding on **September 9, 2026 at 11:59 PM BST**:

Milestone	Phase & Key Tasks	Active Period	Hard Deadline	Key Deliverables
Milestone 1	Draft Report & Pipeline Tag	Aug 25 – Aug 31, 2026	Aug 31, 2026 (11:59 PM BST)	Tag v1.0-peer-review + Draft PDF
Milestone 2	Peer Review on GitHub Issues	Sep 01 – Sep 04, 2026	Sep 04, 2026 (11:59 PM BST)	Post Review on Assigned Group Issue
Milestone 3	Rebuttal & Final Submission	Sep 05 – Sep 09, 2026	Sep 09, 2026 (11:59 PM BST)	Tag v2.0-final + Final Report PDF

2. Biological Background & Context

Cocaine is a potent psychostimulant that inhibits monoamine reuptake (dopamine, serotonin, octopamine/norepinephrine), leading to heightened synaptic signaling, altered locomotion, and drug-seeking behaviors. Understanding how distinct cell populations in the central nervous system respond to acute drug exposure is a central question in neurobiology.

In this study (Bainton et al., *Genome Research*), single-cell RNA sequencing was conducted on *Drosophila melanogaster* brains following acute consumption of sucrose (control) or sucrose supplemented with cocaine. The study investigated sexual dimorphism in behavioral and transcriptional responses across female and male flies in duplicate.

3. Primary Paper Reading Guide

The primary reference paper for this project is provided in your folder: [paper.pdf](#) (GEO: GSE152495).

Before and during your computational analysis, read [paper.pdf](#) with particular attention to:

Key Reading Focus & Guided Questions

Experimental Design (Abstract & Introduction)

- What behavioral assays were performed on flies following acute cocaine exposure?
- Why was single-cell transcriptomics necessary instead of bulk RNA-seq to understand the neural mechanisms of cocaine response?

Computational Methods & Clustering Target

- What clustering algorithm and resolution parameter did the authors use to arrive at **36 distinct cell clusters**?
- How did the authors handle normalization (sequencing depth normalization, SCTransform/regressing UMI counts)?

Figures to Compare with Your Results

- **Figure 2:** UMAP embeddings and cell-type annotations for the 36 clusters. How closely does your Scanpy Leiden clustering (at resolution 0.8) match the major neuronal and glial clusters?
- **Figure 3:** Heatmaps, Venn/Euler diagrams, and volcano plots showing differentially expressed genes in response to cocaine.
- **Figure 4 & 5:** Sexual dimorphism analysis. Notice the difference in the number and magnitude of DE genes in males vs. females. Does your subgroup DE analysis reflect this profound male bias?

Supplementary Material

- Review the supplementary table of marker genes for Kenyon cells, cholinergic neurons, dopaminergic neurons, and glial subtypes to corroborate your findings.

4. Academic Integrity & AI Usage Policy

Generative AI tools (e.g., ChatGPT, Claude, Gemini, GitHub Copilot) may be used as **learning and programming aids** during this project, subject to the following mandatory transparency rules:

Mandatory AI Usage Guidelines

1. Transparency & Prompt Sharing

- If you use AI to write code, debug errors, assist in pipeline design, or interpret biological results, you **MUST include an “AI Usage Disclosure / Appendix”** in your final report.
- In this appendix, you must document:

- The AI tool and model used (e.g., *ChatGPT-4o*, *Claude 3.5 Sonnet*).
- The exact **prompts** you submitted.
- The purpose of each prompt (e.g., *“Formatting rank_genes_groups output into a publication table”* or *“Interpreting Drosophila dopamine pathway genes”*).

2. Critical Verification & Authorship

- AI models frequently hallucinate gene names, pathway annotations, and mathematical formulas. You are solely responsible for validating all computational outputs and biological conclusions.
- Uncritical pasting of AI-generated prose or code constitutes academic misconduct.

3. Code Understanding

- You must be fully prepared to explain all lines of your code and parameter choices during grading or presentation.

5. Environment Management with uv (Fast & Reproducible)

We use **uv** (an extremely fast, modern Python package and project manager written in Rust) instead of Conda. **uv** installs packages in seconds and ensures 100% reproducible environments.

1. Installing uv

- **Linux / macOS / WSL:**

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

- **Windows (PowerShell):**

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

- **Via pip:**

```
pip install uv
```

2. Initializing the Project Environment

In your project repository directory:

```
# 1. Create a virtual environment with Python 3.11
uv venv --python 3.11

# 2. Activate the virtual environment:
# On Linux/macOS/WSL:
source .venv/bin/activate
# On Windows PowerShell:
.venv\Scripts\Activate.ps1

# 3. Install core single-cell packages instantly with uv:
uv pip install scanpy anndata pandas numpy scipy matplotlib seaborn gseapy jupyter ipykernel
```

3. Locking Dependencies for Team Collaboration

Generate a pinned lockfile or requirements file so your team members and peer reviewers can reproduce your exact environment in seconds:

```
uv pip freeze > requirements.txt
```

When team members clone your repository, they can recreate your exact environment instantly:

```
git clone https://github.com/YourUsername/sc-project2-drosophila-brain.git
cd sc-project2-drosophila-brain
uv venv
source .venv/bin/activate # (.venv\Scripts\Activate.ps1 on Windows)
uv pip install -r requirements.txt
```

Alternatively, you can run scripts directly without manual activation using `uv run`:

```
uv run workflow.py
```

6. Research Questions to Address in Your Report

1. **Atlas Construction:** Can you reproduce the **36 distinct brain cell clusters** corresponding to major neuronal, glial, and neurotransmitter-specific subtypes?
2. **Cluster-Specific Cocaine Responses:** Which brain cell types (e.g., Kenyon cells of the mushroom body, glial subtypes, dopaminergic neurons) show the strongest transcriptional changes upon cocaine exposure?
3. **Sexual Dimorphism:** Do male and female flies exhibit differences in the magnitude and identity of cocaine-induced differentially expressed genes?
4. **Pathway Remodeling:** Which biological pathways (e.g., synaptic transmission, metabolic remodeling, circadian rhythm, stress response) are enriched among cocaine-modulated genes?

7. Dataset Overview

Inside your project folder `project2_drosophila_brain/`:

```
data/
├─ Female_Cocaine_1/    (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Female_Cocaine_2/    (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Female_Sucrose_1/    (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Female_Sucrose_2/    (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Male_Cocaine_1/      (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Male_Cocaine_2/      (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
├─ Male_Sucrose_1/      (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
└─ Male_Sucrose_2/      (barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz)
```

Experimental variables: - **Sex:** Female, Male - **Treatment:** Cocaine, Sucrose - **Replicate:** 1, 2

8. Step-by-Step Python Analysis Walkthrough

Step 8.1: Loading Libraries & 10x Data into Scanpy

```
import scanpy as sc
import anndata as ad
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```

from pathlib import Path

# Configure plotting parameters
sc.settings.verbosity = 3
sc.settings.set_figure_params(dpi=120, facecolor="white", frameon=False)

```

Load all 8 sample folders into a unified AnnData object:

```

data_dir = Path("data")

sample_dirs = sorted([
    p for p in data_dir.iterdir()
    if p.is_dir() and (p / "matrix.mtx.gz").exists()
])

print(f"Found {len(sample_dirs)} samples: {[p.name for p in sample_dirs]}")

adatas = {}
for sample_path in sample_dirs:
    sample_name = sample_path.name

    # Load 10x matrix directly
    adata_sample = sc.read_10x_mtx(
        path=sample_path,
        var_names="gene_symbols",
        cache=False
    )

    # Extract metadata: e.g. "Female_Cocaine_1" -> Sex: Female, Treatment: Cocaine, Rep: 1
    parts = sample_name.split("_")
    sex = parts[0]
    treatment = parts[1]
    replicate = parts[2]

    adata_sample.obs["sample"] = sample_name
    adata_sample.obs["sex"] = sex
    adata_sample.obs["treatment"] = treatment
    adata_sample.obs["replicate"] = replicate
    adata_sample.obs["condition"] = f"{sex}_{treatment}"

    adatas[sample_name] = adata_sample

# Concatenate into one object
adata = ad.concat(adatas, label="sample_batch", index_unique="-", join="outer")
adata.var_names_make_unique()

print(f"Merged Dataset: {adata.n_obs} cells x {adata.n_vars} genes")

```

Step 8.2: Quality Control (QC) & Filtering

In *Drosophila melanogaster*: - **Mitochondrial genes**: prefixed with `mt:` (e.g., `mt:CoI`, `mt:ND1`, `mt:Cyt-b`) - **Ribosomal genes**: prefixed with `RpS` or `RpL`

```
# 1. Annotate QC gene groups
adata.var["mt"] = adata.var_names.str.startswith("mt:")
adata.var["ribo"] = adata.var_names.str.startswith(("RpS", "RpL", "rps", "rpl"))

# 2. Compute QC metrics
sc.pp.calculate_qc_metrics(
    adata,
    qc_vars=["mt", "ribo"],
    percent_top=None,
    loglp=False,
    inplace=True
)

# 3. Plot QC violin distributions
sc.pl.violin(
    adata,
    ["n_genes_by_counts", "total_counts", "pct_counts_mt", "pct_counts_ribo"],
    groupby="condition",
    jitter=0.4,
    multi_panel=True
)

# 4. Filter cells and genes
sc.pp.filter_cells(adata, min_genes=200)
sc.pp.filter_genes(adata, min_cells=3)
adata = adata[adata.obs["pct_counts_mt"] < 10.0, :].copy()

print(f"Cells remaining after QC: {adata.n_obs}")
```

Step 8.3: Normalization & Scaling

```
adata.layers["counts"] = adata.X.copy()

# Normalize counts to 10,000 per cell and log-transform
sc.pp.normalize_total(adata, target_sum=1e4)
sc.pp.loglp(adata)
adata.raw = adata

# Select 2,000 highly variable genes
sc.pp.highly_variable_genes(
    adata,
    n_top_genes=2000,
    batch_key="sample"
)
sc.pl.highly_variable_genes(adata)

# Regress out UMI counts and mitochondrial variation
```

```
sc.pp.regress_out(adata, ["total_counts", "pct_counts_mt"])
sc.pp.scale(adata, max_value=10)
```

Step 8.4: Dimensionality Reduction & Replicating the 36 Clusters

The publication found that graph-based clustering with a **resolution of 0.8** yielded **36 distinct clusters**:

```
# PCA
sc.tl.pca(adata, svd_solver="arpack", use_highly_variable=True)
sc.pl.pca_variance_ratio(adata, log=True, n_pcs=50)

# Nearest neighbors & UMAP
sc.pp.neighbors(adata, n_neighbors=15, n_pcs=30)
sc.tl.umap(adata)

# Leiden clustering at resolution 0.8
sc.tl.leiden(adata, resolution=0.8, key_added="leiden_res_0.8")

num_clusters = len(adata.obs["leiden_res_0.8"].unique())
print(f"Identified {num_clusters} clusters (Target: ~36).")

# Visualize UMAP
sc.pl.umap(adata, color=["leiden_res_0.8", "sex", "treatment", "condition"], ncols=2)
```

Step 8.5: Cluster Annotation & Canonical Neurotransmitter Markers

Cell Type / Subtype	Canonical <i>Drosophila</i> Markers	Function / Location
Glial Cells	repo, alrm, moody, wrapper	Surface, cortex, and ensheathing glia
Pan-Neuronal	elav, nSyb, brp	General mature neurons
Kenyon Cells (Mushroom Body)	ey, Fas2, rut, dnc, mb247	Olfactory learning and drug-reward center
Cholinergic Neurons	VAcHt, ChAT	Primary excitatory neurotransmission
GABAergic Neurons	Gad1, VGAT	Primary inhibitory neurotransmission
Glutamatergic Neurons	VGlut	Excitatory neurotransmission / motor circuits
Dopaminergic Neurons	p1e (Tyrosine hydroxylase), DAT	Reward and cocaine sensitization
Serotonergic Neurons	SerT, Trh	Mood and behavioral modulation
Octopaminergic Neurons	Tdc2, Tbh	Invertebrate adrenergic homolog

```
# 1. Calculate cluster markers
sc.tl.rank_genes_groups(adata, groupby="leiden_res_0.8", method="wilcoxon")

# 2. Plot key neurotransmitter and cell-type markers
canonical_markers = ["repo", "elav", "ey", "Fas2", "VACHT", "Gad1", "VGlut", "ple",
"SerT", "Tdc2"]
available_markers = [g for g in canonical_markers if g in adata.var_names]

sc.pl.dotplot(adata, var_names=available_markers, groupby="leiden_res_0.8")
sc.pl.umap(adata, color=available_markers, ncols=3)
```

Step 8.6: Cocaine vs. Sucrose Differential Expression & Sexual Dimorphism

Perform differential expression using the publication's threshold: $|\log_2 FC| > 1.0$ and adjusted $p < 0.05$.

```
# 1. Differential expression in Males (Cocaine vs. Sucrose)
adata_male = adata[adata.obs["sex"] == "Male"].copy()
sc.tl.rank_genes_groups(adata_male, groupby="treatment", reference="Sucrose",
method="wilcoxon")
male_de = sc.get.rank_genes_groups_df(adata_male, group="Cocaine")
male_sig = male_de[(male_de["logfoldchanges"].abs() > 1.0) & (male_de["pvals_adj"] <
0.05)]

# 2. Differential expression in Females (Cocaine vs. Sucrose)
adata_female = adata[adata.obs["sex"] == "Female"].copy()
sc.tl.rank_genes_groups(adata_female, groupby="treatment", reference="Sucrose",
method="wilcoxon")
female_de = sc.get.rank_genes_groups_df(adata_female, group="Cocaine")
female_sig = female_de[(female_de["logfoldchanges"].abs() > 1.0) &
(female_de["pvals_adj"] < 0.05)]

print(f"Number of significant DE genes in Males: {len(male_sig)}")
print(f"Number of significant DE genes in Females: {len(female_sig)}")
```

Step 8.7: Pathway Enrichment Analysis with GSEAPy

```
import gseapy as gp

# Run pathway enrichment on Male cocaine-responsive genes
if len(male_sig) > 0:
    top_male_genes = male_sig["names"].tolist()[:150]
    enr_male = gp.enrichr(
        gene_list=top_male_genes,
        gene_sets=["GO_Biological_Process_2023", "KEGG_2021_Human"],
        organism="fly",
        outdir=None
    )
    print("Top enriched biological pathways in Males:")
    print(enr_male.results.head(10)[["Gene_set", "Term", "Adjusted P-value", "Genes"]])
```

9. Report Structure & Expected Deliverables

1. **Title & Abstract:** Concise summary of the project goals, experimental design, and key findings.
2. **Introduction:** Cocaine neurobiology in *Drosophila*, mushroom body anatomy, and sexual dimorphism.
3. **Methods:** Scanpy workflow details (QC filtering, normalization, PCA, UMAP, Leiden clustering, Wilcoxon test parameters).
4. **Results:**
 - **Figure 1:** QC metrics across sex and treatments.
 - **Figure 2:** UMAP visualization reproducing the ~36 clusters.
 - **Figure 3:** Dotplot/Heatmap identifying neuronal and glial subtypes.
 - **Figure 4:** Volcano plots / tables comparing Cocaine vs. Sucrose in males and females.
 - **Figure 5:** Pathway enrichment bar charts for drug-responsive gene sets.
5. **Discussion:** Interpretation of sexual dimorphism in transcriptional responses and specific vulnerability of Kenyon cells and glia. Direct comparison with findings in [paper.pdf](#).
6. **References:** Cited papers.
7. **Appendix: AI Usage Disclosure:** Full list of AI tools used, exact prompts submitted, purpose, and validation notes.